Question:
Find the largest possible real part of \[(75+117i)z+\frac{96+144i}{z}\]where z is a complex number with $|z|=4$.

Reasoning + Answer:

<think> To find the largest possible real part of the expression \((75+117i)z + \frac{96+144i}{z}\) where \(z\) is a complex number with \(|z|=4\), we can use the properties of complex numbers and the modulus. The real part of the expression is maximized when \(z\) is chosen such that the real part of the expression is maximized. We can use the fact that the magnitude of the expression can be maximized when \(z\) is a suitable complex number that aligns with the terms in the expression.

First, let's consider \(z = 4e^{i\theta}\), where \(\theta\) is a real number, and \(|z| = 4\). Then, we can write:
\[
(75+117i)z + \frac{96+144i}{z} = (75+117i)4e^{i\theta} + \frac{96+144i}{4e^{i\theta}} = 4(75+117i)e^{i\theta} + \frac{24+36i}{e^{i\theta}} = 4(75\cos\theta + 75i\sin\theta + 117i\cos\theta - 117\sin\theta) + (24\cos\theta - 24i\sin\theta) + (36\cos\theta + 36i\sin\theta).
\]
We can separate the real and imaginary parts:
\[
4(75\cos\theta + 117\sin\theta) + 4(117\cos\theta - 75\sin\theta) + 24\cos\theta + 36\cos\theta = 4(75\cos\theta + 117\sin\theta + 117\cos\theta - 75\sin\theta) + (24 + 36)\cos\theta = 4(192\cos\theta + 42\sin\theta) + 60\cos\theta.
\]
To maximize the real part, we need to maximize the expression \(4(192\cos\theta + 42\sin\theta) + 60\cos\theta\). The maximum value of \(192\cos\theta + 42\sin\theta\) is the amplitude of this expression, which is \(\sqrt{192^2 + 42^2} = \sqrt{36864 + 1764} = \sqrt{38628} = 62\sqrt{10}\).

Therefore, the largest possible real part of the expression is:
\[
4 \times 62\sqrt{10} + 60 = 248\sqrt{10} + 60.
\]
However, this is too large for the correct setup. We need to use a more straightforward approach by leveraging the properties of complex numbers and the modulus. Instead, we can use the method of Lagrange multipliers or direct substitution to find the maximum real part, but the key insight is the alignment with the terms. The correct approach shows that the maximum real part is when the terms are aligned appropriately, and the maximum real part is:
\[
\text{Re}\left((75+117i)4 + \frac{96+144i}{4}\right) = \text{Re}(300 + 468i + 24 + 36i) = 324.
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